SAMPLE

Earth's Sky: Solstices and Equinoxes

The Turning Points of the Year

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Introduction

Twice a year, everywhere on Earth except the poles themselves, day and night draw close to equal; and twice a year, the gap between them stretches to its greatest. These four days, the two equinoxes and the two solstices, are the stations of the sun's annual journey north and south across the sky, the turning points of the whole seasonal year, and they belong to the very oldest layer of human astronomy: the solstice alignments of the megaliths and the temples are among the oldest science-bearing structures on the planet, and the equinoxes still run the world's calendars, fixing the date of Easter in the Christian year and the new year in the Persian one, holding the festivals of the harvest and the midwinter, and giving every schoolchild in the temperate zones the first astronomical fact they learn, that the seasons turn on the sun's height.

The subject of this book is the four stations themselves: what they are, why they happen, how the species learned to mark them, and what they have meant. The physics is the tilt of the Earth's axis, the twenty-three-and-a-half-degree lean that makes the sun stand overhead at different latitudes through the year, and the chapter on the science unpacks it plainly; the history runs from the gnomons of the ancient world, whose shadow-lengths recorded the sun's stations for the calendar-keepers, to the discovery of precession, the slow drift of the equinox along the zodiac that Hipparchus caught two thousand years ago and that modern astronomy measures to the fraction of a second; the record runs through the monuments, Stonehenge and Karnak and the Maya cities, whose alignments still work on the appointed mornings; and the modern account covers the refinements, the refraction that makes the equinox day not quite equal, the lag that puts the hottest weather weeks after the solstice, and the leap-year machinery that keeps the calendar's stations pinned to the sky.

A note on the book's method and its neighbors, before the year's circuit begins. The account keeps to the established record: the geometry of the tilt and the orbit, the history of the measurements from the gnomon to the ephemeris, the monuments and calendars as the archaeology and the documentary sources state them, and the modern refinements as the almanacs' own fine print. Two neighboring books in the series hold subjects the stations touch, and this one keeps to its own ground: the calendar's machinery is described here only as it pins the four dates, and the sun's figure-eight across the year's noons, the analemma of the other volume, is mentioned only where the stations' story requires it. The reader who wants the deeper treatment of either neighbor will find it on that neighbor's own pages, and the reader who wants the stations whole holds them here: the tilt, the walk, the shadow, the year, and the four turning days that all of it turns on.

The four days deserve a whole book because they are where astronomy, calendar, climate, and culture intersect: the solstice is an astronomical event, a calendrical festival, a climatic turning point, and a human ritual, all at once, and has been for as long as there are records of what humans did. The chapters that follow move from the discovery of the stations to their science, their observation, their monuments, their calendars, and their place in the present world, and the reader closing this book will see the year itself with new eyes: not a smooth circle of days but a structure, built by the tilt of a planet and read by every culture that ever farmed, sailed, or held a festival under the turning sun.

Chapter 1: The Discovery of Solstices and Equinoxes

The discovery, in the sense of the first human knowing, is beyond record, but its shape can be read from what the early cultures left. The solstice is the sky's most visible cycle: the noon shadow, the sun's rising and setting points on the horizon, and the day's own length all swing through their year, the shadow long at midwinter, short at midsummer, the sunrise point marching along the horizon month by month, and any settled people who watched the sky at all, which is every settled people, saw the swing and marked its ends. The discovery's first form is therefore the horizon mark: the notches and posts and standing stones aligned on the sunrise or sunset at the year's extremes, the device of the megalith builders, and the reader should hold its meaning, that the solstice was the first thing ever instrumented, the first natural event caught in a built structure that lets a group know, on any morning, where the sun's journey stands.

The written discovery, the stations named and computed, belongs to the ancient astronomers of the Near East and the Mediterranean. The Babylonians, from their long series of observations, knew the year's structure and the sun's stations, their texts naming the days of the standing sun; the Egyptian calendar, tied to the Nile and the star Sothis, tracked the year's turning and the temple alignments of Karnak caught the summer solstice sunrise deep in the hall of the great temple; and the Greeks, the first to treat the stations as geometry, gave them their lasting names: the equinox, the equal-night day, when the sun stands upon the celestial equator and the day divides

evenly; and the solstice, the standing-still day, when the sun's rising point pauses at its horizon extreme before turning back, the name recording the observation itself, the sun's standstill, in a word.

The Greek discovery was also the measured one, and the measurement's history carries the subject's first precision. The gnomon, the vertical rod and its shadow, was the standard instrument: the shadow's noon length, tracked through the year, gives the sun's altitude, and its longest and shortest marks are the solstices, the shadow-noon calendar of the Chinese astronomers and the Mediterranean both. Meton's cycle, the nineteen-year pattern that realigned the lunar months with the solar year, was built on the stations' dates; Euctemon and the Greek calendar-makers fixed the year's seasons by the equinoxes; Eratosthenes measured the Earth's circumference using the solstice sun overhead at Syene and the gnomon shadow at Alexandria, the tilt of the planet revealed by the difference; and Hipparchus, comparing the equinox dates of his own era with the records of a century and a half before, found the date drifting, the discovery of precession, the slow westward walk of the equinox point through the zodiac, which is the subject's deepest finding and one of the greatest in ancient astronomy.

A further thread of the discovery deserves its paragraph, the record's own reach into the pre-monumental dark. The solstice-marking habit is older than the megaliths: the painted caves' possible seasonal marks, the carved bone tallies of the ice-age, the Aboriginal Australian stone arrangements with their solstice sightlines, and the oral traditions' star-season calendars of the Pacific and the arctic all carry the watching into the unrecorded past, the year's turning observed by every human range the species has tried. The thread's lesson is the discovery's own universality: the four stations require no instrument but a horizon and a memory, and no people ever lacked them; the megaliths are not the beginning of the solstice-watching but its first permanent record, and the true first discovery is as old as the human attention itself, the day some early watcher of the horizon saw the sunrise point stop, and turn, and knew what the turning meant.

The discovery's cultural record deserves one more entry, the naming's own history. The Latin names arrived with the geometry: solstitium, the standing sun, recording the horizon's pause the watchers saw; aequinoctium, the equal night, recording the balance of the day's division; and the names' translations carried the same facts into the other tongues, the Chinese terms for the extreme and the division, the Sanskrit of the Indian calendrics, the Arabic of the astronomical tradition. The entry's meaning is the subject's universality in the specific: the same two observations, the standstill and the balance, named by every astronomical culture on the planet, the four days the most universally attested facts in the whole human record, known and named wherever the sky was watched and the year was kept.

The written-discovery record of the Chinese tradition deserves its own paragraph, the subject's longest continuous series. The imperial astronomers of China kept the state's year by the gnomon: the official shadow-measurements at the solstices, recorded in the dynastic histories with a continuity of institution and instrument that runs longer than any other scientific series in the world; the calendar-office's year, tuned to the stations and the moons, the leap-months inserted by rule and observation; and the gnomon platforms of the observatories, the state-built instruments whose shadow-tables are the stations' own archive. The tradition's gift to the modern subject is the data: the centuries of solstice shadows, reduced by the modern historians of astronomy, gave the obliquity's historical change one of its longest baselines, the ancient empire's calendar-keeping still serving the science of the present, the longest collaboration in the history of the measurement of the year.

The discovery's modern chapter is the refinement: the exact lean and path of the planet, the obliquity measured to fractions of an arcsecond, the precession measured to the same, the equinox and solstice instants now computed and published years in advance, the stations given to the minute in the almanacs. But the discovery's ancient form, the horizon mark and the gnomon, is still the working instrument of the general reader, and the book's later chapters use it: the four stations of the year are the one part of astronomy that anyone can check, with a stick, a window, and a year's patience, and the reader who does will join the oldest continuous observational tradition the species has, the watching of the sun's turn.